

Muhammad Anas Mansha

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PROFESSIONAL SUMMARY

Software Engineer with 3+ years of experience building scalable backend systems, full-stack web applications, and AI-powered automation solutions. Skilled in TypeScript, Python, NestJS, Next.js, PostgreSQL, Supabase, and Redis, with expertise in backend architecture, REST API development, database design, automation, and scalable production systems.

TECHNICAL SKILLS

Languages: TypeScript, JavaScript, Python, SQL

Frontend: React.js, Next.js, Tailwind CSS, Material UI, Formik

Backend & Databases: NestJS, Node.js, Express.js, Flask, PostgreSQL, Supabase, Redis, REST APIs

Tools & Platforms: Git, GitHub, GitHub Actions, Docker, Linux

Core Competencies: Backend Development, Full-Stack Development, System Design, API Development, Database Design, Authentication & Authorization, AI Integration, Automation, Parallel Processing, Real-Time Data Processing, Production Systems, Agile Development

PROFESSIONAL EXPERIENCE

Software Engineer (Backend / Full Stack)

Feb 2025 – Present

Texagon

- Engineer backend and full-stack applications using NestJS, Next.js, TypeScript, PostgreSQL, Supabase, and Redis for production business platforms.
- Design secure REST APIs and backend services for production applications with emphasis on security, scalability, and maintainability.
- Lead development of AI-powered customer service automation solutions, collaborating directly with stakeholders throughout planning, implementation, and delivery.
- Build scalable backend architecture, authentication systems, database models, and real-time data workflows with emphasis on maintainability and performance.
- Collaborate directly with international clients to gather requirements, propose technical solutions, and deliver production-ready features.
- Contribute to large-scale production systems, delivering scalable backend features and business-critical functionality.

Software Engineer

Mar 2024 – Dec 2024

Magnetar Solutions

- Developed responsive web applications using React.js, Material UI, Formik, and modern JavaScript.
- Improved application architecture through refactoring, modular design, and maintainable coding practices.
- Built Python automation tools for data collection, processing, workflow automation, and pipeline optimization.
- Implemented interactive UI components, animations, and 3D visualizations while ensuring responsive user experience.
- Collaborated within Agile development teams using Git-based workflows and peer code reviews.

Associate Software Engineer

Jul 2023 – Feb 2024

Educative, Inc.

Lahore, Pakistan

- Developed and maintained production features using React.js, JavaScript, Tailwind CSS, and Redis for a large-scale educational platform.
- Collaborated within a large shared codebase, participating in code reviews, debugging, and feature implementation using Git-based workflows.
- Built reusable frontend components while improving application maintainability, consistency, and user experience.
- Applied clean coding principles and Agile development practices to deliver reliable, production-ready software.

Teaching Assistant

Mar 2022 – Jan 2023

Information Technology University

Lahore, Pakistan

- Conducted programming tutorials, mentored students in algorithms and software development, and assisted with grading and coursework.

PROJECTS

Crop Classification Using Satellite Imagery (Final Year Project) | *Python, Machine Learning, Remote Sensing*

- Implemented a machine learning pipeline to classify crop types using Sentinel-2 satellite imagery and vegetation indices, including NDVI.
- Designed scalable geospatial data processing pipelines to efficiently handle large satellite datasets through parallel processing, GeoJSON partitioning, and image stitching.
- Built automated data preprocessing and feature extraction workflows to prepare satellite imagery for model training and inference.
- Collaborated within a multidisciplinary team to develop an end-to-end solution for crop identification and agricultural analysis.

Mass SHL2A Land Processor | *Python, Satellite Imagery, Automation* | GitHub

- Developed a high-performance Python application for automated processing of Sentinel-2 Level-2A satellite imagery.
- Designed a scalable processing pipeline capable of handling large geospatial datasets while reducing manual preprocessing effort.
- Focused on automation, data processing, and workflow optimization for agricultural and land analysis applications.

GTNH Sync | *Python, Desktop Application, Google Drive API* | GitHub

- Created a desktop application that automatically synchronizes the latest Minecraft save-game backups to Google Drive.
- Integrated the Google Drive API to automatically synchronize incremental save-game backups with cloud storage.
- Designed the application to safeguard user data against local storage failures, file corruption, ransomware attacks, and accidental data loss.

Restaurant Management Backend | *NestJS, TypeScript, PostgreSQL* | GitHub

- Designed and implemented scalable REST APIs, authentication, database models, and business logic using NestJS, TypeScript, and PostgreSQL.
- Implemented authentication, business logic, database models, and API endpoints following scalable backend architecture practices.

Bill Split Web Application | *Next.js, React.js* | GitHub

- Built a web application that simplifies shared expense calculations for groups by automatically distributing payment amounts.
- Designed an intuitive interface that eliminated manual calculations for recurring team and group expenses.

EDUCATION

Information Technology University

Bachelor of Science (B.S.) in Computer Science

Lahore, Pakistan

2019 – 2023

ACHIEVEMENTS

- 1st Place – Programming Competition, Superior Technical Symposium
- 1st Place – Taakra Speed Coding Competition, PUCIT
- 1st Place – Speed Programming Competition (Two-Time Winner), University of Lahore
- 2nd Place – Code Rush Programming Competition, Information Technology University
- Selected for STEM NPTC-18 Training Program at Pakistan Institute of Engineering and Applied Sciences (PIEAS)
- Recipient of multiple awards in regional mathematics competitions.